

MCKV INSTITUTE OF ENGINEERING

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Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
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Curriculum for Undergraduate Degree (B.Tech.) in Computer Science and Engineering (AI and ML) (w.e.f. AY: 2022-23)

Part III: Detailed Curriculum

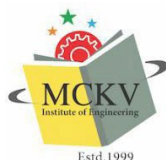
Course Name:	Neural Networks and Deep Learning		
Course Code:	PC-CS(AM)601	Category:	Professional Core Course
Semester:	6th	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	BS-M101, BS-M305
Full Marks:	100		
Examination Scheme:	Semester Examination:70	Continuous Assessment:25	Attendance:05

Course Objectives:

1	To explore the evolution of ANN starting from its initial phase
2	To learn the structure and function of ANN
3	To analyse ANN learning
4	To explore different deep neural networks
5	To learn tools, applications, limitations and future scopes

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Introduction: McCulloch-Pitts Neuron, Perceptron, Perceptron Learning Algorithm and Pattern classification using perceptron, Perceptron function vs Sigmoid or Logistic function, Sigmoid Neuron.	4
2	Structural and Functional framework of ANN: Feedforward Neural Networks: Activation Functions-Hidden Layer, Activation Functions-Output Layer, Multilayer feedforward neural networks with example.	4
3	Learning: Approximation of any arbitrary function, Error or Loss Function- Mean Square and Cross Entropy, Learning Algorithm-Minimization of Loss, Gradient Descent, Backpropagation of error with example, Optimizers, Learning Rate, Overfitting and Underfitting, L1 and L2 Regularization, Dropout, Early Stopping, Augmentation.	8
4	Deep Neural Networks: Convolutional Neural Network (CNN), Pre-Trained Networks, Recurrent Neural Network (RNN) - LSTM, Restricted Boltzmann Machine (RBM), Deep Belief Network (DBN), Autoencoders, Diffusion Model.	8



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5	Generative Adversarial Net (GAN) - Introduction, Applications of GANs, GAN Discriminator, GAN Generator, GAN Training - Upsampling, Transposed Convolutions, Binary Cross Entropy (BCE) Loss for GANs.	6
6	Deep Learning Tools, Research Applications, Limitations of Deep Learning and Potential Future Directions.	6
Total		36L

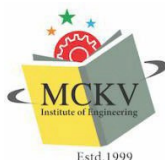
Course Outcomes:

After completion of the course, students will be able to:

1	Explain the concepts of McCulloch-Pitts Neuron, Perceptron and Sigmoid Neuron.
2	Describe the structural and functional framework of Neural Networks.
3	Interpret the learning procedure of Neural Networks.
4	Classify different deep learning models.
5	Recognize different deep learning tools, the research applications, limitations and future directions.

Learning Resources:

1	Neural Networks and Learning Machines, Simon Haykin, Pearson; 3rd edition, 2009.
2	Goodfellow, I., Bengio, Y., and Courville, A., Deep Learning, MIT Press, 2016.
3	Deep Learning: A Practitioner's Approach, Josh Patterson, Adam Gibson, O'Reilly, 2017
4	Fundamentals of Deep Learning, Nithin Buduma, Nikhil Buduma, Joe PapaO'Reilly Media; 2nd edition, 2022.
	Online Resources:
1	https://www.deeplearningbook.org/
2	https://d2l.ai/
3	Modeling Tools: https://www.tensorflow.org/ , https://pytorch.org/ , http://caffe.berkeleyvision.org/ , https://theano-pymc.readthedocs.io/en/latest/



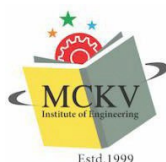
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Course Name:	Database Management System		
Course Code:	PC-CS602	Category:	Professional Core Courses
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic programming Knowledge, Concept of Set theory, Tree Data structure
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To understand the different issues involved in the design and implementation of a database system.
2	To study the physical and logical database designs, database modeling, relational, hierarchical, and network models
3	To understand and use data manipulation language to query, update, and manage a database
4	To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.
5	To understand the different issues involved in the design and implementation of a database system.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to database management, data abstraction and system structure, Database users, Database Administrator	4
2	Entity relational model, entity set, relationship sets, mapping cardinalities, keys, E-R diagrams.	4
3	Relational model, database schema, relational algebra, outer join and manipulation of databases. Tuple relational calculus: Example queries, formal definitions and safety of expressions; SQL: Query processing and optimization, set operations, aggregate functions, data definition language and views, comparison of queries in relational algebra, SQL, tuple relation calculus and domain relation calculus.	8
4	Introduction to Schema Refinement - Problems Caused by redundancy, Decompositions - Problem related to decomposition, Functional Dependencies - Reasoning about FDS, Normal Forms - FIRST, SECOND, THIRD Normal forms - BCNF - Properties of Decompositions - Loss less join Decomposition, Dependency preserving Decomposition, Schema Refinement in Data base Design - Multi valued Dependencies - FOURTH Normal Form, Join Dependencies, FIFTH Normal form, Inclusion Dependencies.	8



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5	Transaction Management, ACID property, Transaction state, Serializability and testing for serializability, concurrency control schemes, lock-based protocols, two-phase locking protocols, graph, time stamp-based protocols, deadlocks.	8
6	Recovery systems, log-based recovery, deferred and immediate database modification, object oriented database design.	2
7	Overview of Storage and Indexing: Data on External Storage, File Organization and Indexing - Clustered Indexes, Primary and Secondary Indexes, Index data Structures - Hash Based Indexing, Tree based Indexing, Comparison of File Organizations.	2
Total		36L

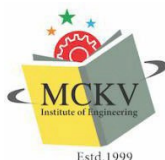
Course Outcomes:

After completion of the course, students will be able to:

1	Describe the fundamental concept of File System and DBMS Architecture.
2	Explain the concepts of different types of attribute, keys and Entity Relationship model.
3	Apply concepts of relational algebra, calculus and Structured Query language.
4	Apply concepts of functional dependency and normalization process to construct normalized database.
5	Implement the isolation property including locking, time stamping based on recovery control and Serializability of schedules.
6	Explain the concepts of Storage and Indexing data, File organization and indexing.

Learning Resources:

1	Data base System Concepts, A.Silberschatz, H.F. Korth, S.Sudarshan, McGraw Hill, VI edition, 2006.
2	Elmasi, R. and Navathe, S.B., "Fundamentals of Database Systems", 4th Ed., Pearson Education
3	Date, C. J., "Introduction to Database Systems", Pearson Education.
4	Ramakrishnan, R. and Gekhre, J., "Database Management Systems", 3rd Ed., McGraw-Hill.



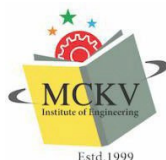
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Course Name:	Cloud Computing		
Course Code:	PC-CS(AM)603	Category:	Professional Core Courses
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Computer Architecture, Operating System
Full Marks:	100		
Examination Scheme:	Semester Examination:70	Continuous Assessment:25	Attendance:05

Course Objectives:	
1	This course gives students an insight into the basics of cloud computing along with virtualization.
2	It will provide the students basic understanding of cloud security and privacy issues.
3	Students will be able to use different cloud services for different purposes.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	<u>Definition of Cloud Computing and its Basics:</u> Defining a Cloud, Cloud Types – NIST model, Cloud Cube model, Deployment models (Public, Private, Hybrid and Community Clouds), Service model: - Infrastructure as a Service, Platform as a Service, Software as a Service with examples of services/ service providers. Cloud Reference model, Characteristics of Cloud Computing, Benefits and advantages of Cloud Computing, A brief introduction on Composability, Infrastructure, Platforms, Virtual Appliances, Communication Protocols, Applications, Connecting to the Cloud by Clients, IaaS –concept, Workload, partitioning of virtual private server instances, Pods, aggregations. SaaS - Basic concept and characteristics, Open SaaS and SOA, examples of SaaS platform Identity as a Service (IDaaS) Compliance as a Service (CaaS)	9
2	<u>Use of Platforms in Cloud Computing:</u> Concepts of Abstraction and Virtualization technologies: Types of virtualization (access, application, CPU, storage), Load Balancing and Virtualization: Basic Concepts, Network resources for load balancing, Advanced load balancing (including Application Delivery Controller and Application Delivery Network), Mention of The Google Cloud as an example of use of load balancing. Hypervisors: Virtual machine technology and types, VMware vSphere Machine Imaging. Porting of applications in the Cloud: The simple Cloud API and AppZero Virtual Application appliance, Concepts of Platform as a Service, Definition of services, Distinction between SaaS and PaaS (knowledge of Salesforce.com and Force.com), Application development Use of PaaS Application frameworks. Discussion of Google Applications Portfolio – Indexed search, Dark Web,	12



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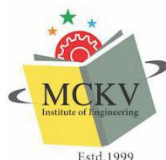
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	Aggregation and disintermediation, Productivity applications and service, Adwords, Google Analytics, Google Translate, a brief discussion on Google Toolkit (including introduction of Google APIs in brief), major features of Google App Engine service. Windows Azure platform: Microsoft's approach, architecture, and main elements, overview of Windows Azure AppFabric, Content Delivery Network, SQL Azure, and Windows Live services.	
3	<u>Cloud Infrastructure:</u> Cloud Management: An overview of the features of network management systems and a brief introduction of related products from large cloud vendors, Monitoring of an entire cloud computing deployment stack – an overview with mention of some products, Lifecycle management of cloud services (six stages of lifecycle). Concepts of Cloud Security: Cloud security concerns, Security boundary, Security service boundary Overview of security mapping Security of data: Brokered cloud storage access, Storage location and tenancy, encryption, and auditing and compliance Identity management (awareness of Identity protocol standards).	7
4	<u>Concepts of Services and Applications:</u> Service Oriented Architecture: Basic concepts of message-based transactions, Protocol stack for an SOA architecture, Event-driven SOA, Enterprise Service Bus, Service catalogs, Applications in the Cloud: Concepts of cloud transactions, functionality mapping, Application attributes, Cloud service attributes, System abstraction and Cloud Bursting, Applications and Cloud APIs Cloud-based Storage: Cloud storage definition – Manned and Unmanned Webmail Services: Cloud mail services including Google Gmail, Mail2Web, Windows Live Hotmail, Yahoo mail, concepts of Syndication services.	8
Total		36L

Course Outcomes:

After completion of the course, students will be able to:

1	Explain the fundamental concept of cloud computing and its characteristics, benefits and limitations.
2	Classify different types of cloud models, architecture and infrastructure of cloud computing and its examples.
3	Demonstrate abstraction and different types of virtualization, load balancing technology and their role in the cloud computing model.
4	Solve the security, privacy issues and cloud management of cloud computing.
5	Use various cloud services in different applications.



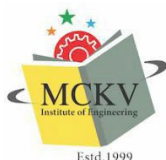
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Learning Resources:	
1	Cloud Computing Bible by Barrie Sosinsky, Wiley India Pvt. Ltd, 2013
2	Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, McGraw Hill Education (India) Private Limited, 2013
3	Cloud computing: A practical approach, Anthony T. Velte, Tata Mcgraw-Hill
4	Cloud Computing, Miller, Pearson
5	Building applications in cloud: Concept, Patterns and Projects, Moyer, Pearson
6	Cloud Computing – Second Edition by Dr. Kumar Saurabh, Wiley India
7	Cloud Computing – Second Edition by Dr. Kumar Saurabh, Wiley India



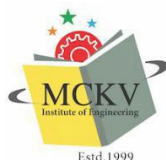
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Course Name:	Data Mining		
Course Code:	PE-CS601A	Category:	Professional Elective
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic concepts of Database, Concept of Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To identify the scope and essentiality of Data Mining and Warehousing
2	To analyze data, choose relevant models and algorithms for respective applications.
3	To develop research interest towards advances in data mining.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Data Mining overview, Data Warehouse and OLAP technology, Data Warehouse Architecture, Steps for the Design and Construction of Data Warehouses, A Three – tier Data Warehouse Architecture, OLAP, OLAP queries, metadata repository, Data Preprocessing, Data – Integration and Transformation, Data Reduction, Data Mining Primitives: What defines a Data Mining Task? Task Relevant Data, The Kind of Knowledge to be mined, KDD.	06
2	Mining Association rules in Large Databases, Association Rule Mining, Market Basket Analysis: Mining a Road Map, The Apriori Algorithm: Finding Frequent Itemsets using Candidate Generation, Generating Association Rules from Frequent Itemset, Improving the Efficiency of Apriori, Mining Frequent Itemset without Candidate Generation, Multilevel Association Rules, Approaches to Mining Multilevel Association Rules, Mining Multidimensional Association Rules for Relational Databases and Data Warehouses, Multidimensional Association Rules, Mining Quantitative Association Rules, Mining Distance Based Association Rules, From Association Mining to Correlation Analysis.	10
3	Classification and Prediction, Issues regarding Classification and Prediction, Classification by Decision Tree Induction, Bayesian Classification, Bayes Theorem, Naïve Bayesian Classification. Classification based on Concepts from Association Rule Mining and Other Classification methods like k-Nearest Neighbor Classifiers, Prediction, Linear and Multiple Regression, Non-Linear Regression, Other Regression Models, Classifier Accuracy.	10



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4	Cluster Analysis, Data types in Cluster Analysis, Categorization of Major Clustering Methods, Classical Partitioning Methods: k-Means and k-Medoids, Partitioning Methods in Large Databases i.e k-Medoids to CLARANS, Hierarchical Methods, Agglomerative and Divisive Hierarchical Clustering, Density-Based Methods, CLIQUE: Clustering High-Dimensional Space, Model Based Clustering Methods.	10
Total		36L

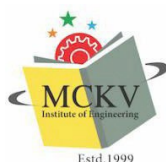
Course Outcomes:

After completion of the course, students will be able to:

1	Identify the key processes of data mining, data warehousing and knowledge discovery process.
2	Implement association rule mining to solve real world problems.
3	Implement different data mining techniques for classification and regression analysis.
4	Compare different clustering algorithms in relevance to high dimensional space.

Learning Resources:

1	Jiawei Han and M Kamber, Data Mining Concepts and Techniques,, Second Edition, Elsevier Publication, 2011
2	Vipin Kumar, Pang-Ning Tan, Michael Steinbach, Introduction to Data Mining - Addison Wesley,2006.
3	Arun K. Pujari, Data Mining Techniques, Universities Press, 2001
4	Alex Berson and Stephen J. Smith, Data Warehousing, Data Mining, & OLAP, Second Edition Tata McGraw Hill Education
5	Pang-Ning Tan, "Introduction to Data Mining", Addison Wesley, 2006.



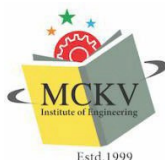
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Course Name:	Pattern Recognition		
Course Code:	PE-CS 601B	Category:	Professional Elective Courses
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Discrete Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To introduce students to the application of the Bayesian Decision theorem for pattern classification
2	To make the students realize the need for dimensionality reduction and choosing a proper dimensionality reduction model depending on the problem statement
3	To introduce students to different classification and clustering methods depending on the nature of the pattern

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Pattern; Feature; Feature Selection; Dimensionality; Types of Learning	2
2	Basics of Probability: Probability; Independence of events, conditional and joint probability, Bayes' theorem; Distribution Functions; Bayes Decision Theory	2
3	Bayes Decision Theory: Minimum-error-rate classification, Classifiers, Discriminant functions, Decision surfaces, Normal density and discriminant functions, discrete features, Naïve Bayesian Classifier	5
4	Parameter Estimation Methods: Maximum-Likelihood estimation: Gaussian case; Maximum a Posteriori estimation; Bayesian estimation: Gaussian case Nonparametric techniques for density estimation: Histogram Based Method, Windows Based Methods, Parzen-window method; K-Nearest Neighbour method	5
5	Dimensionality reduction: Introduction, Problems of High dimensionality, Principal component analysis; Linear Discriminant Analysis	4
6	Sequential Pattern Recognition: Hidden Markov Models (HMMs); Discrete HMMs	4
7	Linear discriminant functions: Gradient descent procedures; Perceptron; Support vector machines (brief introduction)	6



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	Non-linear discriminant functions: Introduction to Multilayer Perceptron; Brief introduction to Back Propagation Algorithm	
8	Non-metric methods for pattern classification: Non-numeric data or nominal data; Decision trees: ID3	4
9	Unsupervised learning and clustering: Criterion functions for clustering; Algorithms for clustering: K-Means, Hierarchical clustering, Self-Organizing Map	4
Total		36L

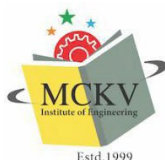
Course Outcomes:

After completion of the course, students will be able to:

1	Demonstrate the concept of Bayesian Decision Theory for the classification of data having parametric and non-parametric PDF.
2	Compare different dimensionality reduction methods.
3	Solve problems related to sequential data using the Discrete Hidden Markov Model.
4	Differentiate problems related to linear classifiers and non-linear classifiers.
5	Solve problems related to non-metric methods of pattern classifications.
6	Demonstrate the concept of clustering.

Learning Resources:

1	R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2001
2	S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009
3	C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006



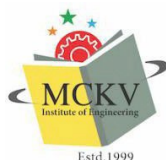
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Course Name:	Image Processing		
Course Code:	PE-CS601C	Category:	Professional Elective
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Mathematics I (BS-M-101) Mathematics II (BS-M-201) Computer Graphics (PE-CS501C)
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To understand the fundamentals of digital image processing.
2	To familiarize with the basic image model suitable for computer processing
3	To familiarize with mathematical preliminaries of digital image processing.
4	To understand different image enhancement methods.
5	To understand different image restoration procedures.
6	To understand applications of different image processing techniques in image segmentation

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Digital Images, Fundamental steps in Digital Image Processing, Components of Digital Image Processing system. Elements of Digital Image Processing - Image Acquisition, Storage, Processing, Communication, Display.	3
2	Image Model: Image model suitable for computer processing, Sampling & Quantization - Uniform & Non uniform.	3
3	Mathematical Preliminaries: Neighborhood, Connectivity, boundaries, Relations, Distance Measures, Arithmetic/Logic Operations. Complex Numbers, Fourier Transformation and its properties. Two Dimensional Fourier Transform and Discrete Fourier Transform basic and applications.	6
4	Image Enhancement: Spatial Domain Method, Frequency Domain Method, Basic Intensity Transformation: Image negative, Log transform, Gamma transform, Piecewise linear transform, Histogram Processing: Histogram equalization, Histogram specification, Global and local histogram processing, Histogram statistics	8



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	Spatial Filtering: Correlation and Convolution, Spatial filter mask, Smoothing spatial filters-Linear and Non-linear, sharpening Spatial filters-Gradient, Laplacian, Unsharp masking, Highboost filtering.	
5	Image Restoration: Degradation Model, Discrete Formulation, Differences in noise removal and restoration, noise models, Different spatial and frequency domain filters, Estimation of degradation function, Inverse filters; Wiener Filtering, Geometric Transformation - Spatial Transformation, Gray Level Interpolation.	8
6	Image Segmentation: Point, Line and Edge detection, Gradient operator, Combined detection, Edge Linking & Boundary Detection - Local Processing, Global Processing via Hough Transform; Thresholding – Otsu method, Single and multiple threshold, Variable thresholding, Region Oriented Segmentation - Region Growing, Region Splitting & Merging.	8
Total		36L

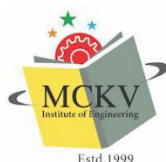
Course Outcomes:

After completion of the course, students will be able to:

1	Explain the basic concepts of digital image fundamentals and computer processing of image models.
2	Discuss different image enhancement techniques.
3	Interpret Fourier transform for image processing in frequency domain.
4	Interpret different image processing techniques for image restoration and segmentation.

Learning Resources:

1	Digital Image Processing by Richard E. Woods and Rafael C. Gonzalez, Pearson
2	Digital Image Processing and Analysis by Chanda Bhabatosh, Majumder Dutta Dwijesh, PHI.
3	Digital Image Processing, Castleman, Pearson
4	Fundamentals of Digital Image Processing, A. K. Jain, PHI



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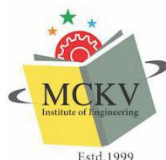
Course Name:	Distributed Computing		
Course Code:	PE-CS602A	Category:	Professional Elective
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Java Programming, Operating Systems, Computer Networks
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To introduce the fundamental concepts and issues of managing large volume of shared data in a parallel and distributed environment, and to provide insight into related research problems.
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Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Fundamentals: Evolution of Distributed Computing Systems, System models, issues in the design of Distributed Systems, Distributed computing environment, web-based distributed model, computer networks related to distributed systems and web-based protocols.	6
2	Message Passing: Interprocess Communication, Desirable Features of Good Message-Passing Systems, Issues in IPC by Message, Synchronization, Buffering, Multidatagram Messages, Encoding and Decoding of Message Data, Process Addressing, Failure Handling, Group Communication.	6
3	Remote Procedure Calls The RPC Model, Transparency of RPC, Implementing RPC Mechanism, Stub Generation, RPC Messages, Marshaling Arguments and Results, Server Management, Communication Protocols for RPCs, Complicated RPCs, Client-Server Binding, Exception Handling, Security, Some Special Types of RPCs, Lightweight RPC, and Optimization for Better Performance.	6
4	Synchronization Clock Synchronization, Event Ordering, Mutual Exclusion, Election Algorithms.	3
5	Resource and Process Management Desirable Features of a good global scheduling algorithm, Task assignment approach, Load Balancing approach, Load Sharing Approach, Process Migration, Threads, Processor allocation, Real time distributed Systems.	6
6	Distributed File Systems Desirable Features of a good Distributed File Systems, File Models, File Accessing Models, File-sharing Semantics, File caching Schemes, File Replication, Fault Tolerance, Design	6



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	Principles, Sun's network file system, Andrews file system, comparison of NFS and AFS	
7	Introduction to parallel and cluster computing.	3
Total		36

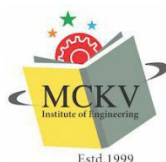
Course Outcomes:

After completion of the course, students will be able to:

1	Demonstrate knowledge of the basic elements and concepts related to distributed system technologies.
2	Illustrate the middleware technologies that support distributed applications such as RPC, RMI and Object based middleware.
3	Investigate the various techniques used for clock synchronization and mutual exclusion.
4	Demonstrate the concepts of Resource and Process management and synchronization algorithms.
5	Investigate various file systems like NFS, AFS and the experience in building large-scale distributed applications through the knowledge of Distributed File System.

Learning Resources:

1	Andrew S. Tanenbaum and Maarten Van Steen, Distributed Systems: Principles and Paradigms, 2nd edition, Pearson Education.
2	George Coulouris, Jean Dollimore, Tim Kindberg, "Distributed Systems: Concepts and Design", 4th Edition, Pearson Education, 2005.
3	M. L. Liu, Distributed Computing Principles and Applications, Pearson Addison Wesley, 2004.



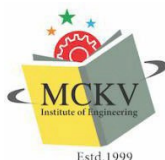
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Course Name:	Soft Computing		
Course Code:	PE-CS602B	Category:	Professional Elective
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Basic knowledge of Set theory
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To familiarize the students with the basic concepts of soft computing and its techniques.
2	To acquaint the students with fuzzy logic and Genetic Algorithm.
3	To develop the ability to perform different types of learning with the help of Artificial Neural networks.

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction to Soft Computing: Introduction to soft computing; introduction to fuzzy sets and fuzzy logic systems; introduction to biological and artificial neural network; introduction to Genetic Algorithm.	2
2	Fuzzy Logic: Introduction, Operations on Classical sets, properties of classical sets, Properties of fuzzy sets, Linguistic Hedges, Fuzzy set operations. Features of membership functions, standard forms and boundaries, different fuzzification methods, Lambda Cuts for fuzzy sets, Defuzzification methods. Crisp Logic, Fuzzy Logic, Fuzzy Rule based Inference System – Mamdani FIS, Sugeno FIS. Applications of Fuzzy Inference System	10
3	Artificial Neural Network: Introduction to Artificial Neural Networks, Biological Neurons and Artificial Neural Network, Resemblance of Biological neuron and ANN, Basic Idea of learning. Basic terminology of ANN, Topology of ANN, Different Activation Function, MP Neural Network. Neural Network for Supervised Learning: Perceptron and Delta learning Rule, ADALINE, Single layer Perceptron, MADALINE, Multilayer Perceptron with Back propagation Algorithm Neural Network for Unsupervised Learning: Hebbian Learning Rule, Competitive ANN like Kohonen's Self Organizing Feature Map. Associative Memory using ANN: Associative memory, Bi-directional Associative Memory, Binary Hopfield Networks. Neuro-Fuzzy modelling: Applications of Neural Networks: Pattern Recognition and classification	14
4	Genetic Algorithms: Basic Idea of Optimization, Introduction to Simple GA, brief introduction to other optimization techniques with Pros and Cons, Simple GA Algorithm GA operator: Different selection techniques, Different non-constrained and	8



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	constrained Crossover and Mutation techniques Introduction to Multi-objective Genetic Algorithm (MOGA), pareto Optimal Solutions Genetic algorithms in search and optimization, GA based clustering Algorithm, Image processing, and Pattern Recognition	
5	Other Soft Computing techniques: Simulated Annealing, Ant colony optimization (ACO), Particle Swarm Optimization (PSO).	2
Total		36L

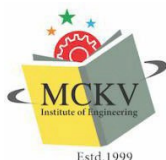
Course Outcomes:

After completion of the course, students will be able to:

1	List the difference between Hard Computing and Soft Computing
2	Implement a Fuzzy Inference System by practising fuzzy set theory and fuzzy logic to illustrate
3	Implement an appropriate Artificial Neural Network for solving a given problem on Supervised learning, Unsupervised learning and Content Addressable Memory
4	Describe the Simple Genetic Algorithm and its operators
5	Recall the other Soft Computing Techniques such as Simulated Annealing, ACO, Swarm Optimization, MOGA

Learning Resources:

1	Principles of Soft Computing, S N Sivanandam, S.N. Deepa, Wiley India
2	S. Rajasekaran and G.A.V.Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithms", PHI
3	Neural Networks: A Classroom Approach, 1/e by Kumar Satish, TMH,
4	Soft Computing and Intelligent System Design, Theory, Tools and Applications, F.O. Karray, C. De Silva, Pearson
5	Genetic Algorithms in search, Optimization & Machine Learning by David E. Goldberg
6	Neuro-Fuzzy and Soft computing, Jang, Sun, Mizutani, PHI
7	Fuzzy logic with engineering applications, Timothy J. Ross, John Wiley and Sons.
8	Genetic Algorithms in search, Optimization & Machine Learning by David E. Goldberg, Pearson/PHI
9	A beginner's approach to Soft Computing, Samir Roy & Udit Chakraborty, Pearson
10	Fuzzy Sets and Fuzzy Logic: Theory and Applications, George J. Klir and Bo Yuan, Prentice Hall
11	Neural Networks: A Comprehensive Foundation (2nd Edition), Simon Haykin, Prentice-Hall.



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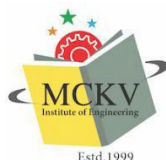
Course Name:	Data Analytics and Visualization		
Course Code:	PE-CS602C	Category:	Professional Elective
Semester:	Sixth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Concept of Data Science and Machine Learning
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:

1	To understand Data Analytics Life Cycle and Business Challenges
2	To understand Analytical Techniques and Statistical Models

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Module 1: Data Analytics Lifecycle Data analytics, Drivers of data analytics, Typical analytical architecture, Need of Data analytic lifecycle, Various phases of Data analytic lifecycle: Discovery, Data Preparation, Model Planning, Model Building, Communicating Results, Operationalization, Key roles for successful analytic projects	6L
2	Module 2: Data Cleaning, Preparation and Visualization Data Cleaning and Preparation: Handling Missing Data - Data Transformation: Removing Duplicates, Transforming Data Using a Function or Mapping, Replacing Values, Detecting and Filtering Outliers- String Manipulation: Vectorized String Functions in pandas.	6L
3	Module 3: Data Visualization Foundation Exploring the Visual Data Spectrum- Charting Primitives (Data Points, Line Charts, PerfChart, Seasonality Chart, Bar Charts, Pie Charts, Donut Charts, Area Charts), Exploring advanced Visualizations (Candlestick Charts, Bubble Charts, Surface Charts, Map Charts, Heat Maps, Gantt Chart, Polygon Maps, Infographics).	6L
4	Module 4: Introduction to Data Visualization Acquiring Data, Visualizing Data, Simultaneous Acquisition and Visualization, Plotting with pandas: Line Plots, Bar Plots, Histograms and Density Plots, Scatter or Point Plots. Applications of Data Visualization (Uses in the Public Sector, Business-to-Business and inter-Business Uses, Business- to-Consumer Uses),	8L



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5	Module 5: Introduction to Tableau Overview, Environment Setup, Design Flow, File Types, Data Types, Tableau data sources: Data Sources, Custom Data View, Extracting Data, Fields Operations, Editing Metadata, Data Joining, Data Blending, Tableau Worksheets: Add Worksheets, Rename Worksheet, Save & Delete Worksheet, Reorder Worksheet, Paged Workbook, Tableau Calculations: Operators, Functions, Numeric Calculations, String Calculations, Date Calculations, Table Calculations, LOD Expressions, Tableau Sort & Filters: Basic Sorting, Basic Filters, Quick Filters, Context Filters, Condition Filters, Top Filters, Filter Operations	10L
Total		36L

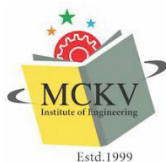
Course Outcomes:

After completion of the course, students will be able to:

1	Prepare the data analytics lifecycle to address data analytics projects
2	Use appropriate analytic techniques and tools to analyse data and create statistical models
3	Prepare appropriate visualizations to clearly communicate analytic insights to business sponsors and analytic audiences
4	Use tools such as Tableau Software in database analytics

Learning Resources:

1	R N Prasad and Seema Acharya, Fundamentals of Business Analytics, Wiley (Latest Edition)
2	Wes McKinney, "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython", O'Reilly, 2 nd Edition, 2018.
3	Daniel G. Murray, Tableau Your Data!: Fast and Easy Visual Analysis with Tableau Software, Wiley (Latest Edition)
4	Jon Raasch, Graham Murray, Vadim Ogievetsky, Joseph Lowery, JavaScript and JQuery for Data analysis and visualization, Wiley (Latest Edition)



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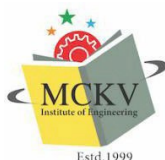
Course Name:	Database Management System Lab		
Course Code:	PC-CS 692	Category:	Professional Core
Semester:	Sixth	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Basic understanding in database management
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:

1	Learn to create and use a database.
2	Be familiarized with a query language.
3	Have a good understanding of DDL, DML and DCL commands.
4	Familiarize advanced SQL queries.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Structured Query Language Creating Database, Creating a Table, Specifying Relational, Data Types Specifying Constraints, Creating Indexes.	3
2	Table and Record Handling INSERT statement, Using SELECT and INSERT together, DELETE, UPDATE, TRUNCATE statements, DROP, ALTER statements.	6
3	Retrieving Data from a Database The SELECT statement, Using the WHERE clause, Using Logical Operators in the WHERE clause, Using IN, BETWEEN, LIKE, ORDER BY, GROUP BY and HAVING Clause *Using Aggregate Functions *Combining Tables Using JOINS * Subqueries	9
4	Database Management Creating Views, Creating Column Aliases, Creating Database ,Users, Using GRANT and REVOKE commands – Commit, Rollback, Save point.	9
5	PL/SQL Concepts Introduction, Cursors, Stored Procedures, Stored Functions, Database Triggers.	9
Total		36P



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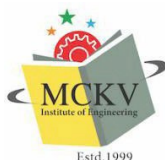
Course Outcomes:

After completion of the course, students will be able to:

1	Implement the concept of Data Definition Language, and Data manipulation language.
2	Use the concept of Column level, row level and check constraints.
3	Implement the concept of Aggregation functions, Joins and views.
4	Implement the concept of Cursors, triggers in PL/SQL.

Learning Resources:

1	Beginning SQL Programming, Kauffman, SPD/WROX
2	Ivan Bayross, "SQL, PL/SQL the Programming Language of Oracle"



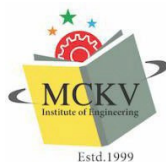
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Course Name:	Software Development and IT Operations Lab		
Course Code:	PC CS 694	Category:	Professional Core
Semester:	Sixth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	The basic idea of the Computer Networks, and Database Management System
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 5

Course Objectives:	
1	To learn the design and development process involved in creating a cloud-based application
2	Able to maintain cloud security.
3	Able to use different web service automation tools

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Identity Access Management, Root Account Vs IAM user, Multi-Factor Authentication for Users, IAM Password Policies, Creating Customer Managed Policies, Glacier Storage, what is Simple Storage Service (S3), Storage Classes, Cross-region replication, Life Cycle Management, Security & Encryption, Static Webhosting with S3 bucket, Events configuration on S3 buckets, Enabling cross-account access for S3, S3 Data management and backup using 3rd Party applications, S3 Cross-Account Access and Pre-Signed URLs Storage Gateway.	2
2	Basics of Linux for AWS, Web Server and Services Configurations, EC2 Instance Launch Wizard, EC2 Instance Types, Generating custom Public Key and Private keys for EC2 instances, Security groups, Volumes and Snapshots, Creating customized Amazon Machine Images, User. Data and Metadata, Elastic Load Balancers & Health Checks, Auto Scaling Groups, CloudWatch, Creating Billing Alarm and EC2 instance alarms, AWS CLI&EC2 Roles, Elastic File System, AWS Lightsail, Elastic Beanstalk Placement Groups.	6
3	DNS Records overview Routing Policies, Hosting sample Website and configuring Policies, Simple Routing Policy, Latency Routing Policy Failover Routing Policy Weighted Routing Policy, Geolocation Routing Policy, Databases, Launching a RDS Instances (MySQL, MSSQL & Aurora), Multi-AZ & Read Replicas for RDS instances, DynamoDB, Redshift, ElastiCache, Database Migration Service and Schema conversion tool.	2
4	VPC (Virtual Private Cloud), Networking Basics, Creating custom VPCs and custom Subnets, Network ACL's, Route Tables & IGW, VPC Peering, Flow log creation VPN Configuration with AWS (OpenVPN). Security Options, CloudTrail, AWS Config, Key Management Services, AWS Certificate Manager, AWS Trusted Advisor Content Delivery Networks / CloudFront.	2
5	Application Services, Simple Email Service, Simple Queue Service, Simple Workflow Service, Simple Notification Service, SMS — Server Migration Service, Migrating server from on-premises to cloud, Cloud Formation, Directory Services and Adding EC2	2



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	instance to Domain, AWS TCO Calculator and Simple Monthly calculator.	
6	DevOps Tools Overview, What is DevOps in Cloud, Code Pipeline, Code Commit, Code Build, Code Deploy Lambda.	8
7	Monitoring Tools, Creating Custom Metrics with CloudWatch, Deploy node project in AWS.	2
Total		24

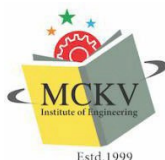
Course Outcomes:

After completion of the course, students will be able to:

1	Implement MFA, IAM user, Budget configuration, private and public bucket in S3 storage, EC2 service.
2	Implement website hosting on S3 and EC2 and also deploy a project on EC2 from GitHub account.
3	Implement Auto Scaling in AWS that balances the load on different EC2 instances.
4	Implement various AWS cloud services like Database (RDS), WorkMail, Elastic IP, Lambda, DevOps, CloudWatch, Security Tools, and Amazon DNS service etc.

Learning Resources:

1	Aws: The Ultimate Guide from Beginners To Advance For The Amazon Web Services (2020 Edition) by Theo H. King.
2	Amazon Web Services in Action by Andreas Wittig, Michael Wittig, Ben Whaley.
3	Aws: Amazon Web Services. The Ultimate Guide for Beginners, Intermediates, and Expert by Phillip Broyles.



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Course Name:	Aptitude Skill Development-II		
Course Code:	MC671	Category:	Mandatory Courses
Semester:	Sixth	Credit:	0
L-T-P:	2-0-0	Pre-Requisites:	Quantitative Ability, Logical and Verbal Reasoning
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		

Course Objectives:	
1	To be prepared in the area of Quantitative Ability as well as Logical and Verbal Reasoning for Campus Placements and different Competitive Exams

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Verbal: Reading Comprehension, Para Jumbles, Email Writing, Resume Writing	3L
2	Game based Cognitive Skills, Tournaments	3L
3	Solve company-oriented campus placements aptitude papers covering Quantitative Ability, Logical Reasoning and Verbal Ability.	12L
4	MCQ Based Strategies/Short cuts and Mock test	6L
Total		24L

Course Outcomes:	
After completion of the course, students will be able to:	
1	Express idea in verbal and non-verbal communication in an effective manner
2	Apply cognitive skills in problem solving
3	Apply reasoning skills in problem solving

Learning Resources:	
1	Arun Sharma, "Quantitative abilities", McGraw-Hill
2	R.S.Agrawal, "Quantitative Aptitude for Competitive Examinations", S. Chand
3	R.S.Agarwal, "A Modern Approach to Verbal & Non-Verbal Reasoning", S.Chand